

applications, integrating multiple regulated power supplies, system management, and display functions in a small package. It consists of three programmable step-down converters and eleven programmable general-purpose LDOs.

Moreover, TPS6521845 and TPS6521835 from TI feature three adjustable step-down converters, one adjustable buck-boost converter, two low-quiescent current, high-efficiency step-down converters for battery backup domain, and an adjustable general-purpose LDO. They are built for battery-powered portable systems as well as 5V non-portable systems.

ROHM Semiconductor BM81810MUF-ME2 and

BM81810MUV-M allow voltage selection via the I2C interface.

Integrated battery charging

Battery-powered devices use PMICs available with integrated battery chargers. Most portable and compact integrated systems are powered by Li-ion/Li-po batteries. A dedicated 2-stage or 3-stage charging system along with a battery monitoring system for battery management is required for such batteries.

Recently launched Nordic Semiconductor's nPM1100, as shown in Fig. 1, comes with an integrated battery charger that can charge both Li-ion and Li-po batteries. It can detect USB port voltage and accordingly power the load. The PMIC aims to maximise the battery life through high efficiency and low quiescent currents.

TI's TPS65951, seen in Fig. 2, also features a Li-ion, Li-po, and cobalt-nickel-manganese charger as it is built for cellular handsets.

Maxim Integrated's MAX77659 features a battery charger with programmable battery regulation voltage, wide current range, and fast-charging current. This flexibility

allows the user to integrate batteries according to their design convenience.

Application based features

Latest PMICs provide additional support for peripherals according to their respective applications. Apart from this, some PMICs also present new application-specific features, like MAX2520 and VR5510.

Maxim Integrated's MAX2520, intended for camera applications, features a soft-start for a smooth transient response. It also supports high-frequency operation for improved noise immunity and load-transient response.

NXP's VR5510, designed for gateway, advanced driver assistance systems (ADAS), vehicle-to-everything (V2X), and infotainment applications, provides external frequency synchronisation on inputs and outputs for optimised system electromagnetic compatibility (EMC) performance.

TI's TPS65921, made for multimedia processors, consists of a real-time clock (RTC), a keypad interface, and embedded power control (EPC). TI's TPS65951 features an LED driver circuit, RTC, external vibrator, etc., as it is intended to be used in cellular handsets. Similarly, TPS658643 of TI PMIC, meant for handheld devices, provides support for dual RGB LED drivers.

Conclusion

Power management for complex integrated systems is always a challenge. Solutions to tackle this will continue to improve with time in terms of efficiency, accuracy, noise immunity, compactness, and performance. PMICs are also becoming more scalable with features like programmability, interface support, wide range of input and output voltages, and high current handling capacities.

In conclusion, we can say power management is quite an emerging technology with significant advancements happening consistently. **EFY**

Darshil Patel worked as a technical journalist at EFY until recently.



Fig. 2: Texas Instruments' TPS65951 (Credit: <https://www.ti.com>)

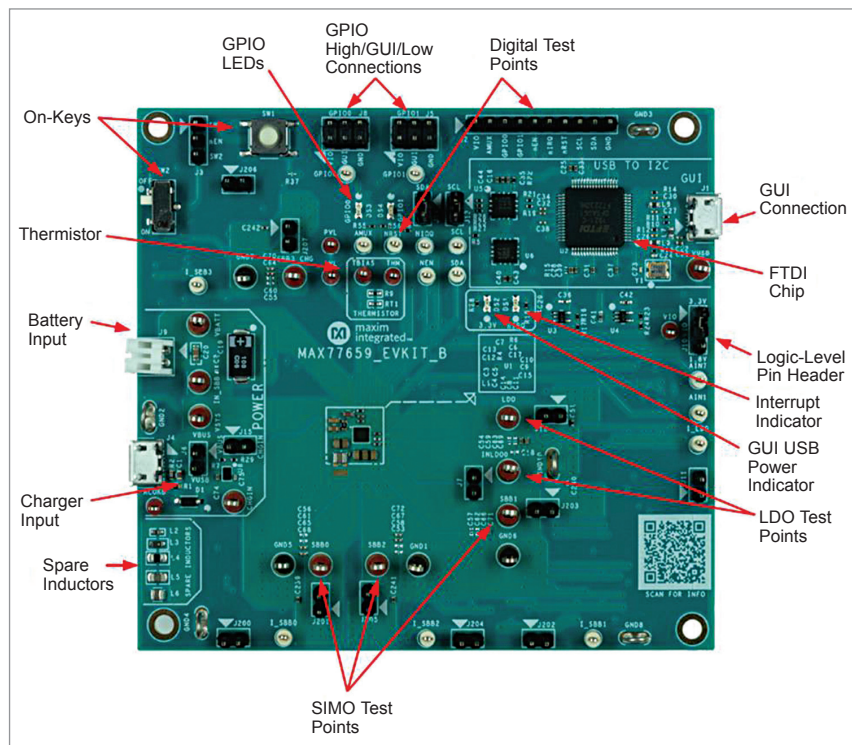


Fig. 3: Maxim Integrated's MAX77659 (Credit: <https://www.mouser.in>)